# JavaScript

Practical Lab Session 3 — Level 2 — Student Worksheet
Practice: Variables, Data Types & Operators

## Part 1 — Operator Precedence: Level Up

**Task 1.1 — Trace It Yourself:** let val = 4 + 3 * 2 ** 2 - 6 / 3; Predict the value by writing out each step (like the guided example above), THEN verify in the console.

**Answer :**

Predicted Result - 14

Step1 - 2**2 = 4
Step2 - 3*4 = 12
Step3 - 6/3 = 2
Step - 4 + 12 = 16
Step - 16 - 2 = 14

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Lab - 3</title>
</head>
<body>
    <h2>JS Lab - 03</h2>
    <br>
    <br>
    <p><b>Q1. </b>4+3*2**2-6/3</p>

    <p>

    </p>
    <p id="result"></p>


    <script src="script.js"></script>
</body>
</html>
```

script.js

```
let val = 4+3*2**2-6/3;

console.log(val)

document.getElementById("result").textContent = "Result - " + val;
```

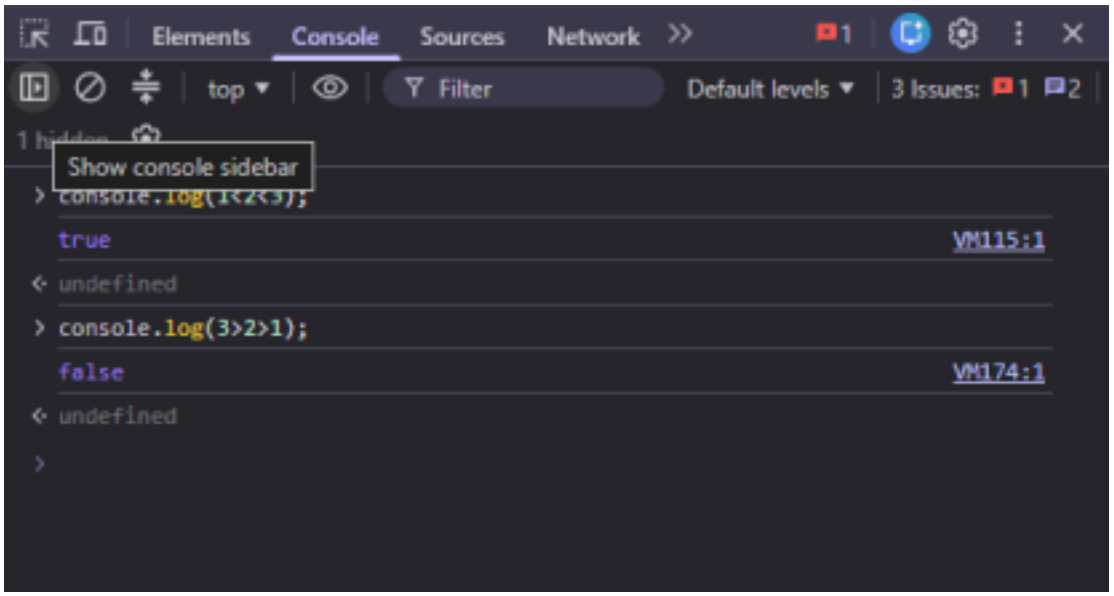



**Task 1.2 — Comparison Chaining Gotcha:** Predict, then run, both lines below. The second resultsurprises most people — explain why in 2–3 sentences, tracing left to right.
console.log(1 < 2 < 3); // predict first
console.log(3 > 2 > 1); // predict first

**Answer :**

1. 1st line will result true because js follows left to right method for comparision.

**2.** 2nd line will result false because when 3>2 = true true=1 , 1>1 is false .



**Task 1.3:** Write ONE expression that combines at least 4 different operators (mixing arithmetic and comparison) and evaluates to true. Show your reasoning, step by step, like the guided example.

**Answer :**

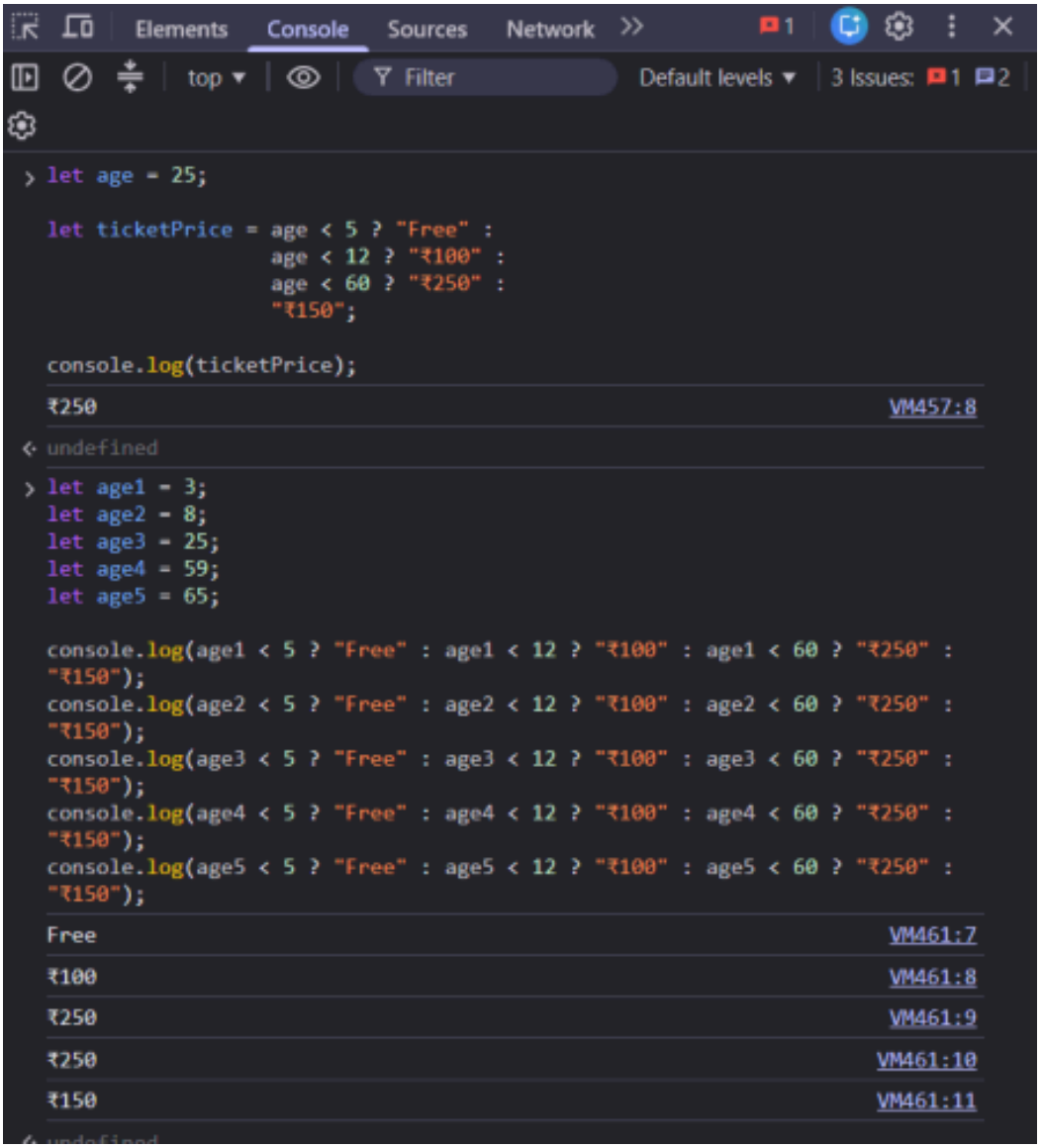
Step by step:

1. Addition: 8+4=12
2. Multiplication: 12×2=24
3. Subtraction: 20−1=19
4. Comparison: 24>19 → True
5.

## Part 2 — Nested Ternary Decision Logic

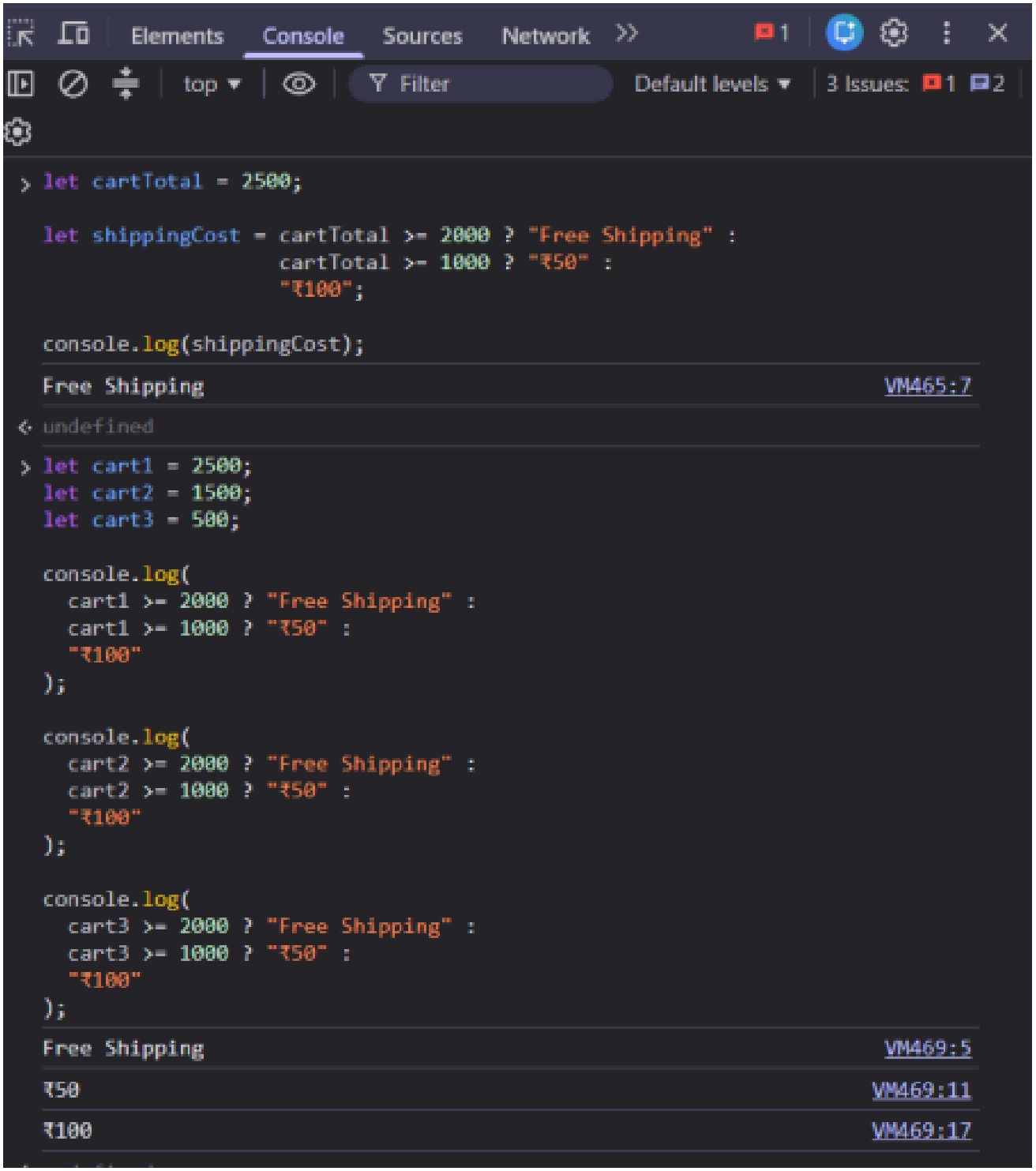
**Task 2.1 —** Movie Ticket Pricing: Build a nested ternary: age < 5 → "Free", age < 12 → "₹100", age < 60 → "₹250", else (senior citizen) → "₹150". Test it with 5 different ages.

**Answer :**



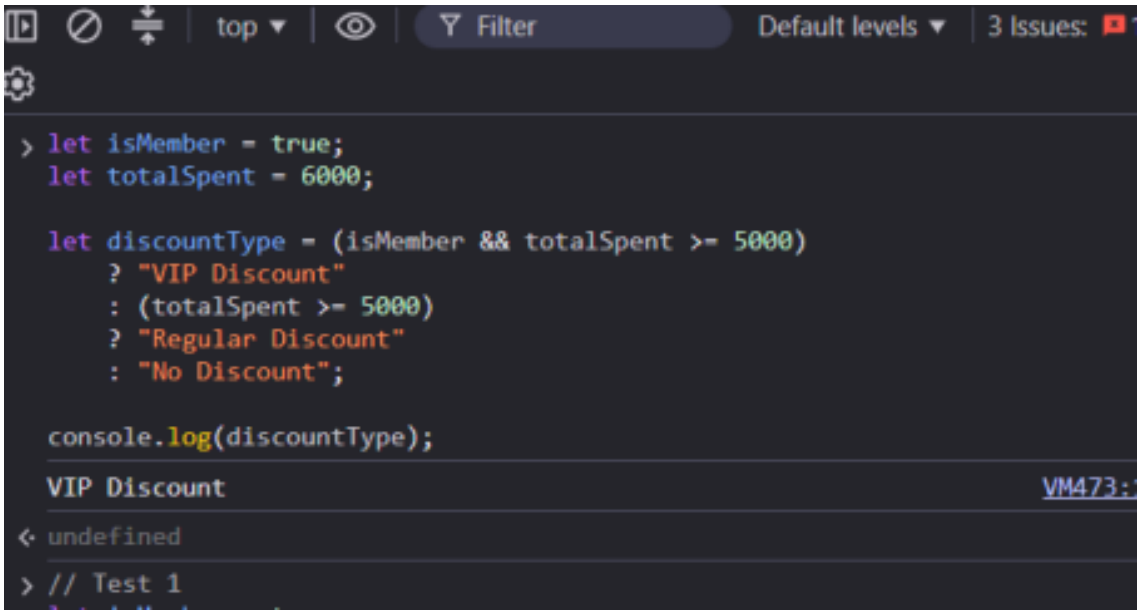
**Task 2.2 —** Shipping Cost Calculator: cartTotal >= 2000 → "Free Shipping", cartTotal >= 1000 → "₹50", else → "₹100". Test with 3 different cart totals.

**Answer :**



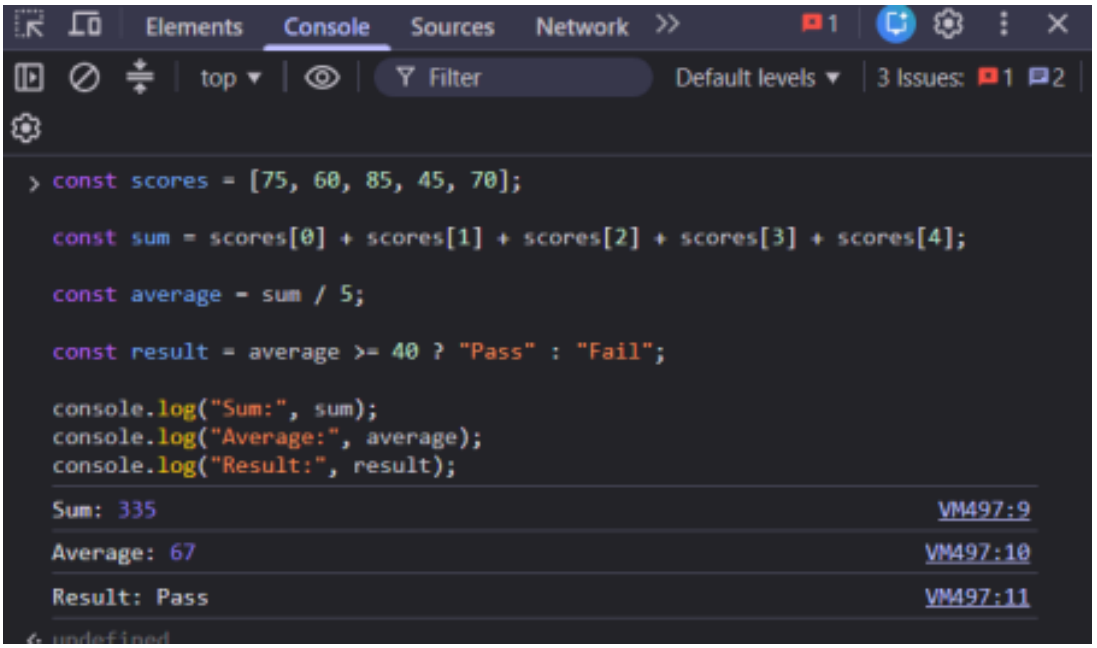
**Task 2.3 —** Trickier: Ternary + Logical Operator Together: Write a SINGLE ternary expression where discountType is "VIP Discount" if (isMember AND totalSpent >= 5000), "Regular Discount" if (totalSpent >= 5000 but not a member), or "No Discount" otherwise. Use && inside the ternary Condition.

**Answer :**



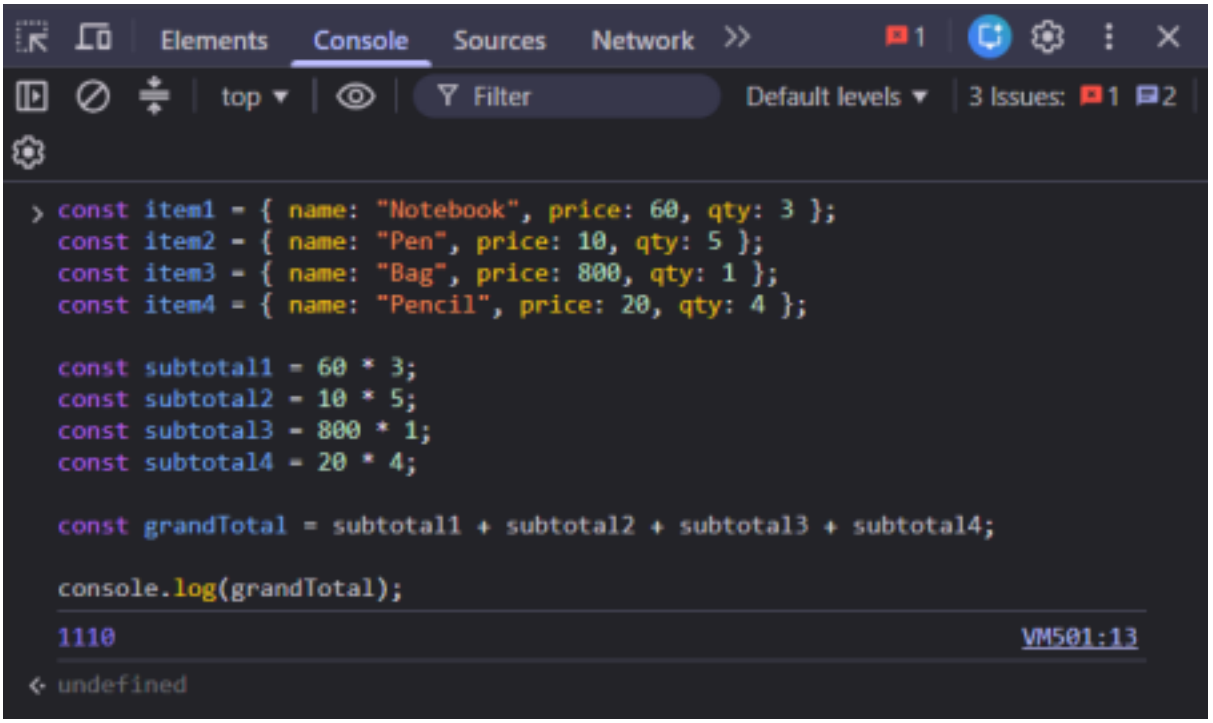

## Part 3 — Manual Aggregation: Arrays & Objects without Loops

**Task 3.1**: Given an array of 5 exam scores of your choice, manually compute the sum and average using indexing. Then use a ternary to classify the average as "Pass" (>= 40) or "Fail".
**Answer :**



**Task 3.2:** Add a 4th product object of your choice, calculate its subtotal, and recompute the grand total by including it.

**Answer :**




## Part 4 — Deep Dive: Tricky Type Coercion

**Task 4.1:** For all 7 lines above, record the actual output and a one-line reason in your lab record. Circle the one you find most surprising and explain why in 2–3 sentences.

**Answer :**

| Expression | My Prediction | Actual Output | One-line Reason |
|---|---|---|---|
| "5" + 3 | "53" | "53" | + with a string performs string concatenation, so 3 becomes "3". |
| "5" - 3 | 2 | 2 | - converts the string "5" to the number 5, then subtracts 3. |
| "abc" * 2 | NaN | NaN | "abc" cannot be converted to a number, so multiplication produces NaN. |

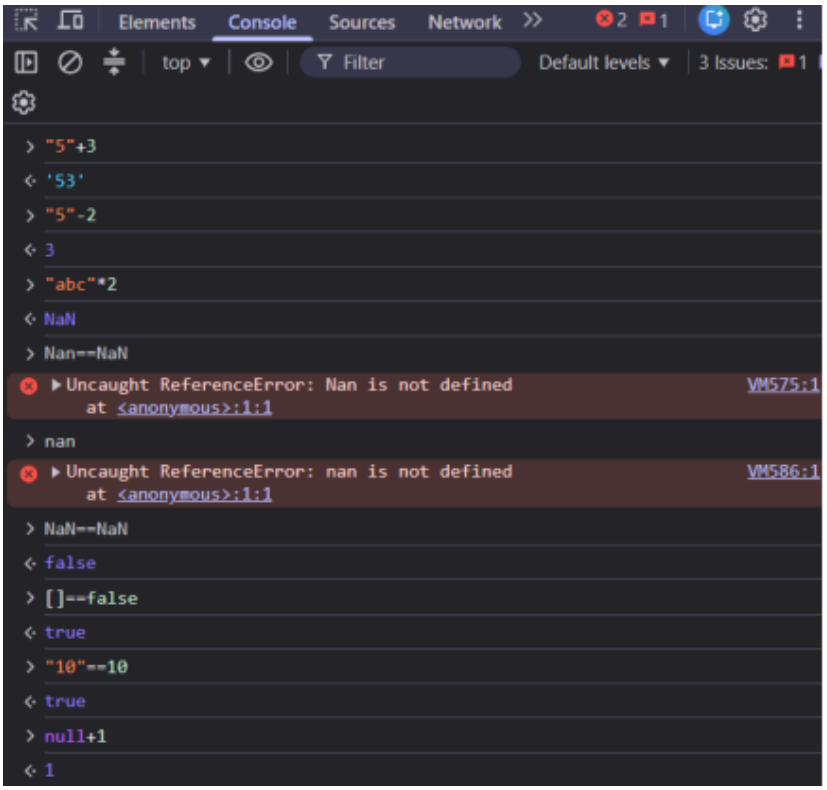
| | | | |
|---|---|---|---|
| NaN === NaN | false | false | `NaN` is not equal to itself, even with strict equality. |
| [] == false | true | true | Loose equality converts `[]` to `""`, then to `0`, which equals `false` after conversion. |
| "10" == 10 | true | true | Loose equality converts the string `"10"` to the number `10`. |
| null + 1 | 1 | 1 | In numeric addition, `null` is converted to `0`, so `0 + 1` is `1`. |

**Task 4.2:** Predict, then run, typeof NaN. Explain why this result seems contradictory given that "NaN" stands for "Not-a-Number."

**Answer :**

Prediction of typeof NaN =number

NaN means "Not a Number", but it is actually a special value inside JavaScript's number data type.



# Part 6 — Debugging Challenge: Multi-Bug Snippets

**Snippet 1**

```
let price = 500;

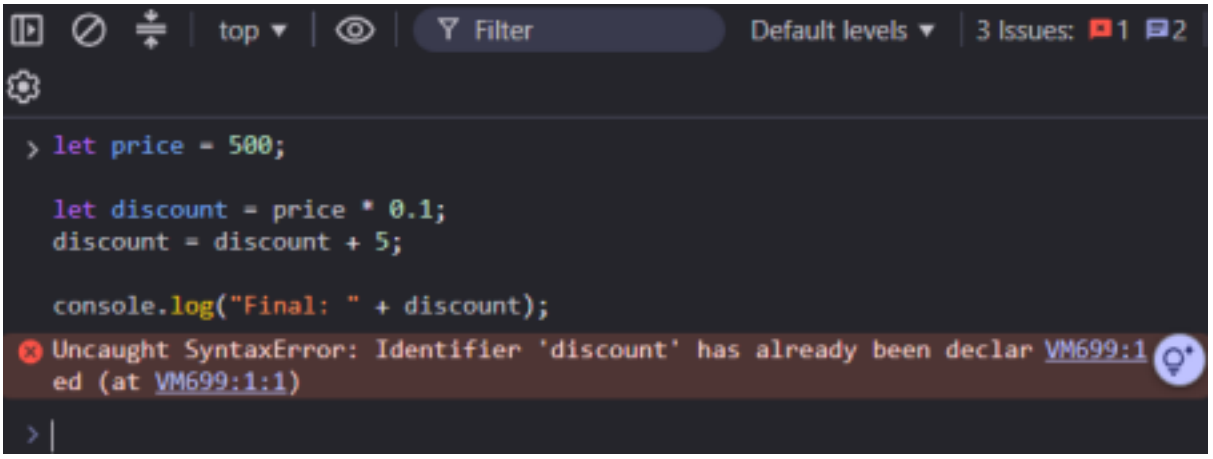
const discount = price * 0.1;

discount = discount + 5;

console.log("Final: " + fnal);
```

**ANSWER:**

Discount is declared using const so it cannot be reassigned that why it is showing error.




**Snippet 2** (silent bug — no crash on the first bug!)

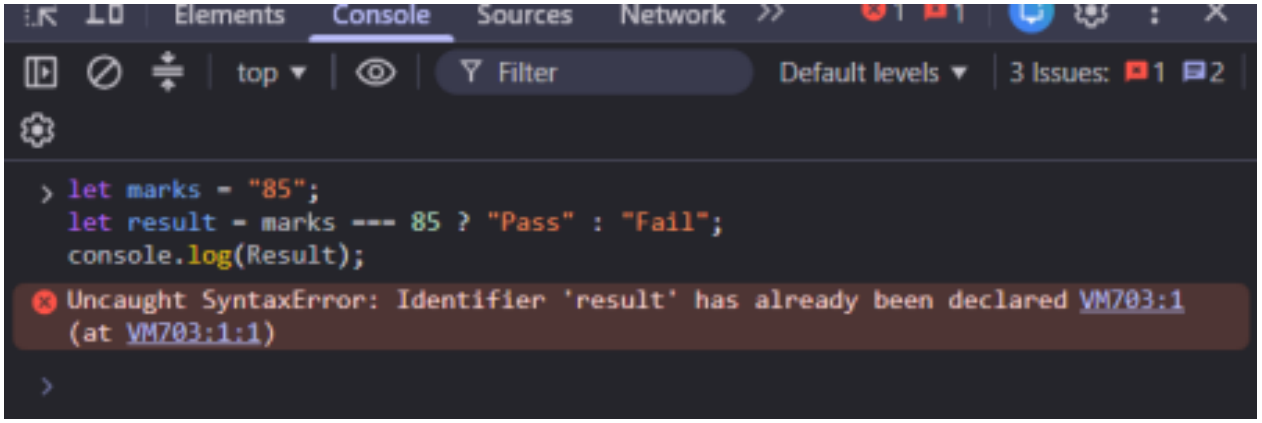
```
let marks = "85";

let result = marks === 85 ? "Pass" : "Fail";

console.log(Result);
```

**ANSWER:**

While printing the result in consol.log thr Rsult is used which is not equal to result==Result.



**Snippet 3**
**Page 4 of 4**

```
const cartTotal = 1200;

cartTotal = cartTotal - 100;

let shipping = cartTotal >= 1500 ? "Free" "Paid";

console.log(shipping);
```

**ANSWER:**

**Const can not be reassigned and : is not present in ternary operator.**

